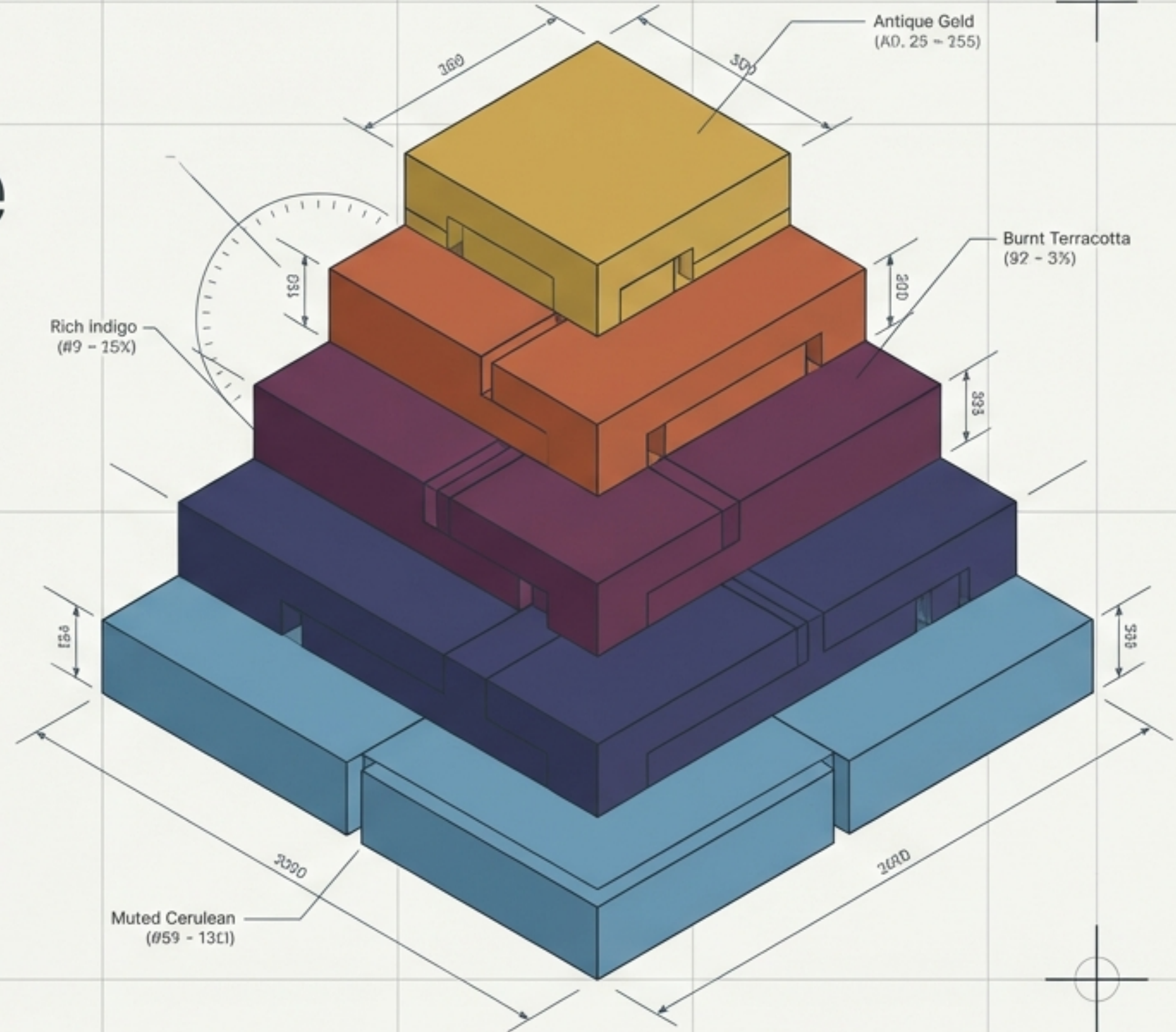


The Architecture of Trust

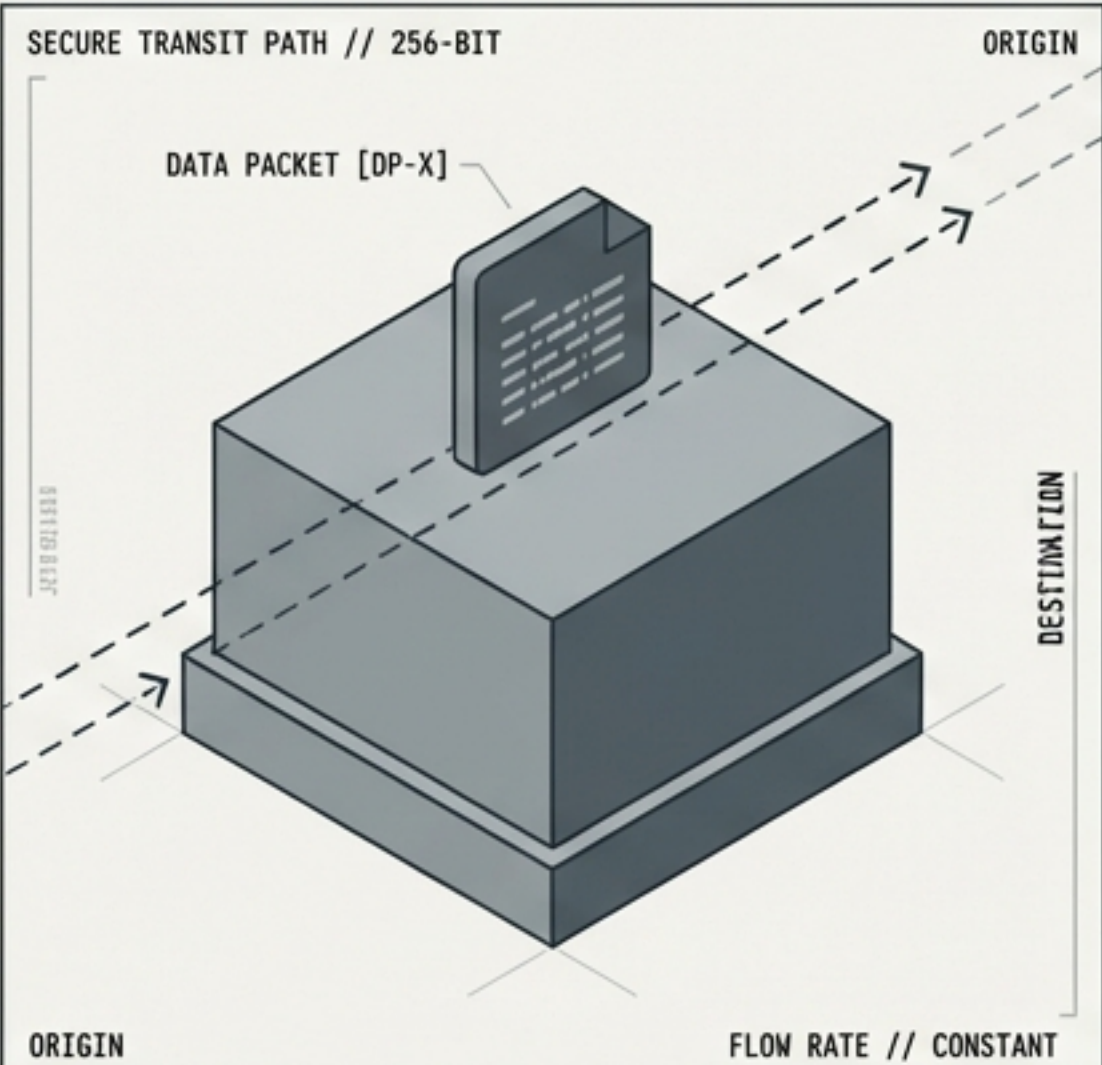
A strategic blueprint for building secure, scalable applications on SG/Send.

Navigating the five tiers of cryptographic complexity, collaboration, and infrastructure.



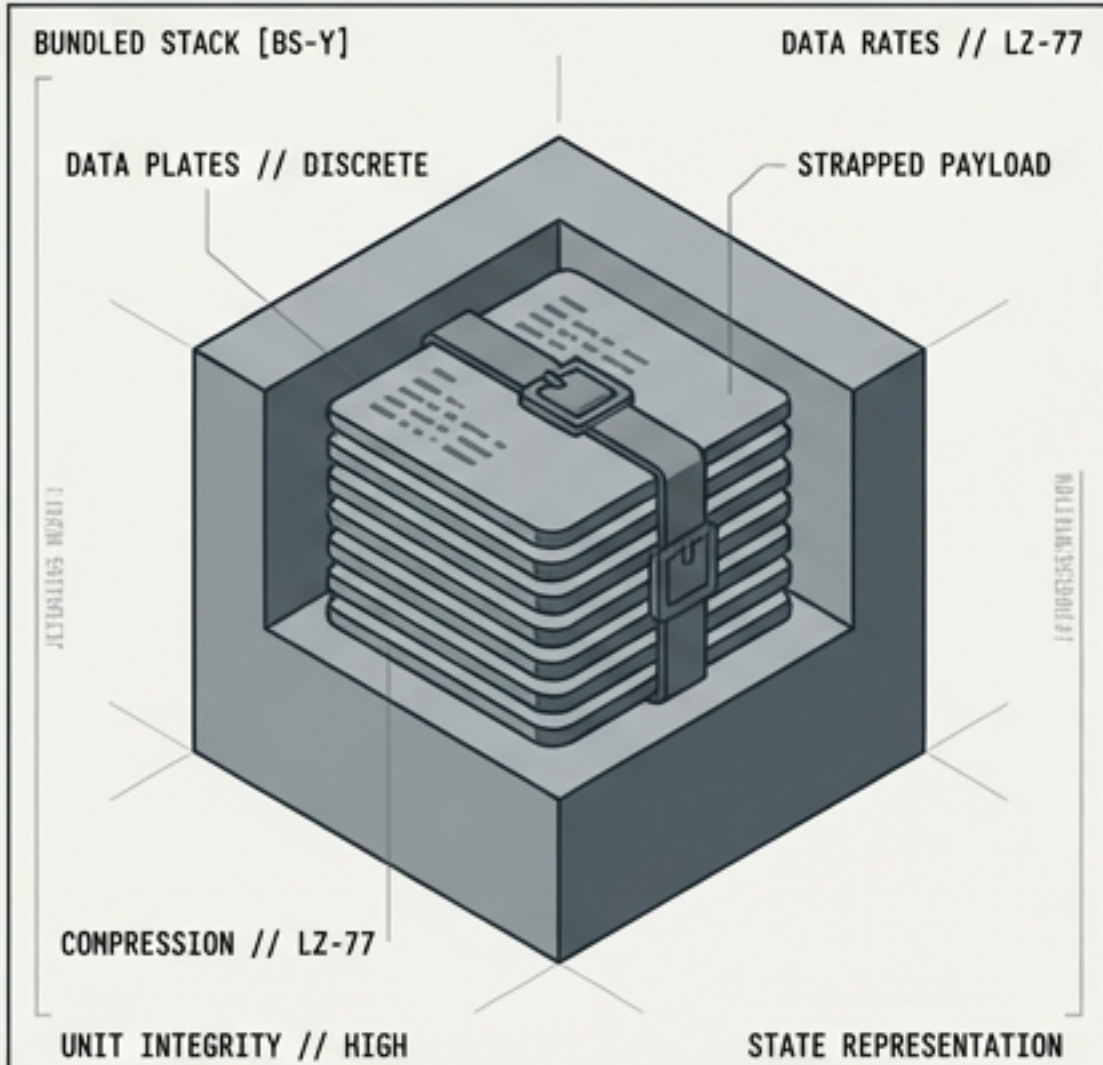
The Three Immutable Primitives

Every use case, from ephemeral messaging to air-gapped critical infrastructure, is built on the same foundation. Different tiers simply lean on different primitives.



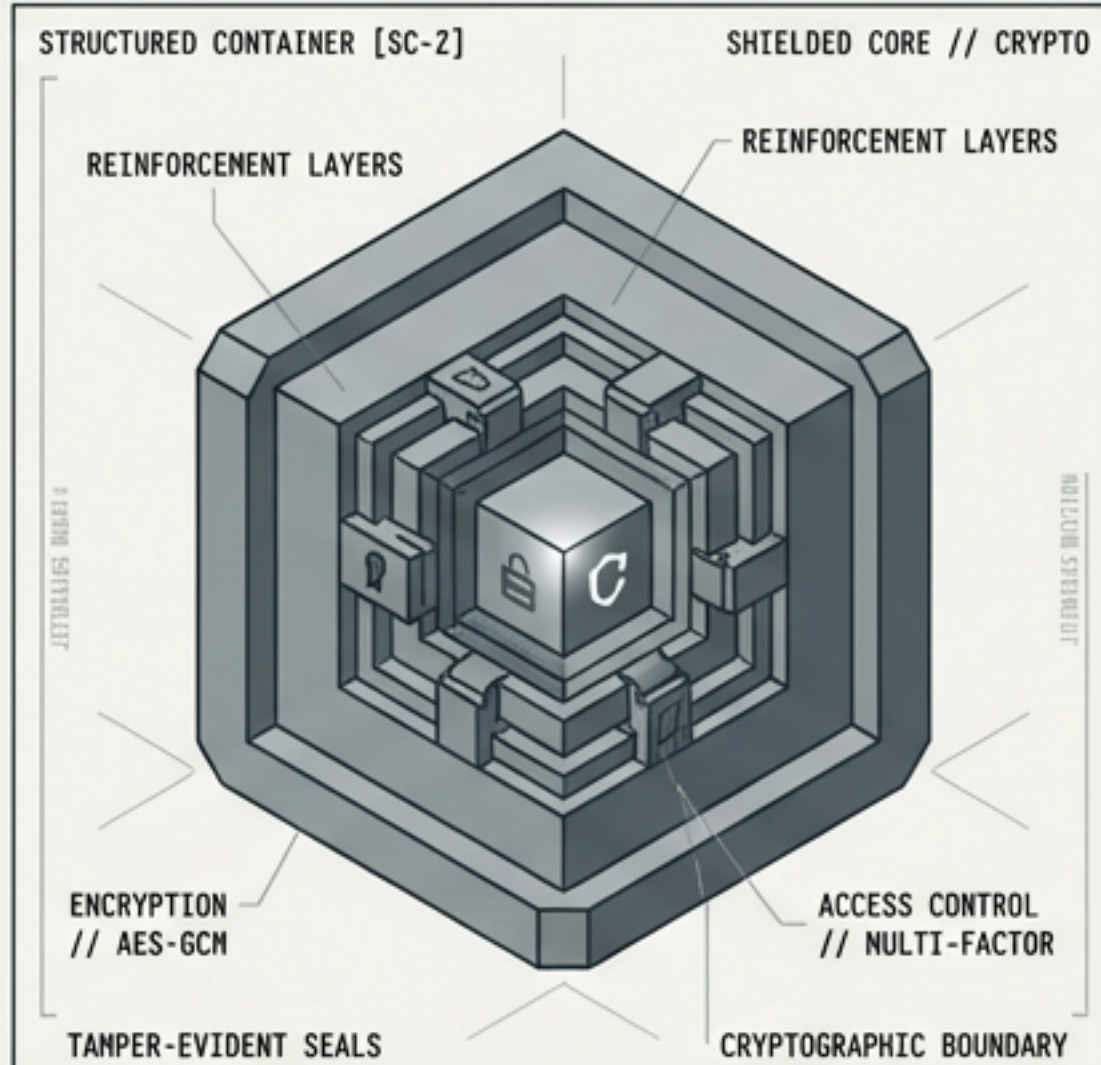
Encrypted File Transfer

Point-to-point secure transit.



Folder / Zip

Bundled state representation.

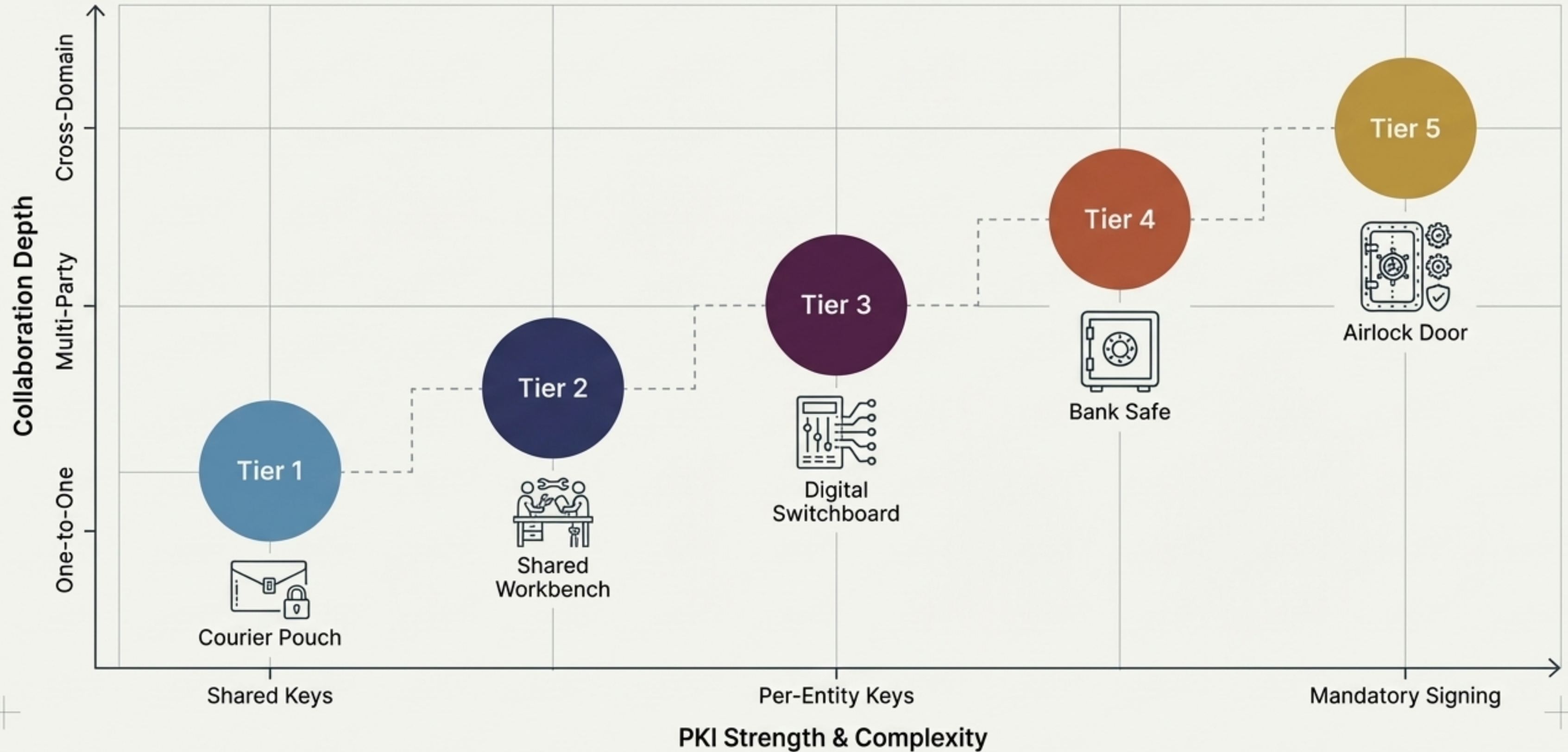


Vault

The structured cryptographic container.

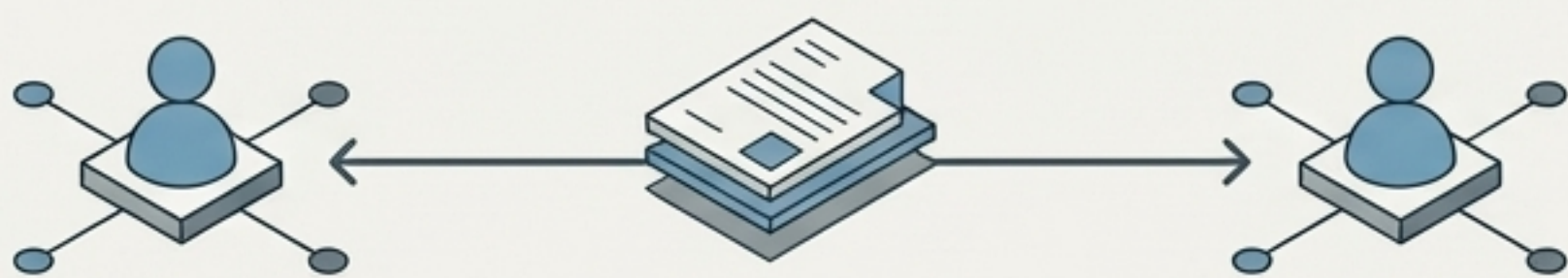
The Progressive Staircase: Mapping the 5 Tiers

Complexity is a **choice**. Choose the tier that matches your domain's regulatory and collaboration demands.



Tier 1: Point-to-Point Secure Transfer

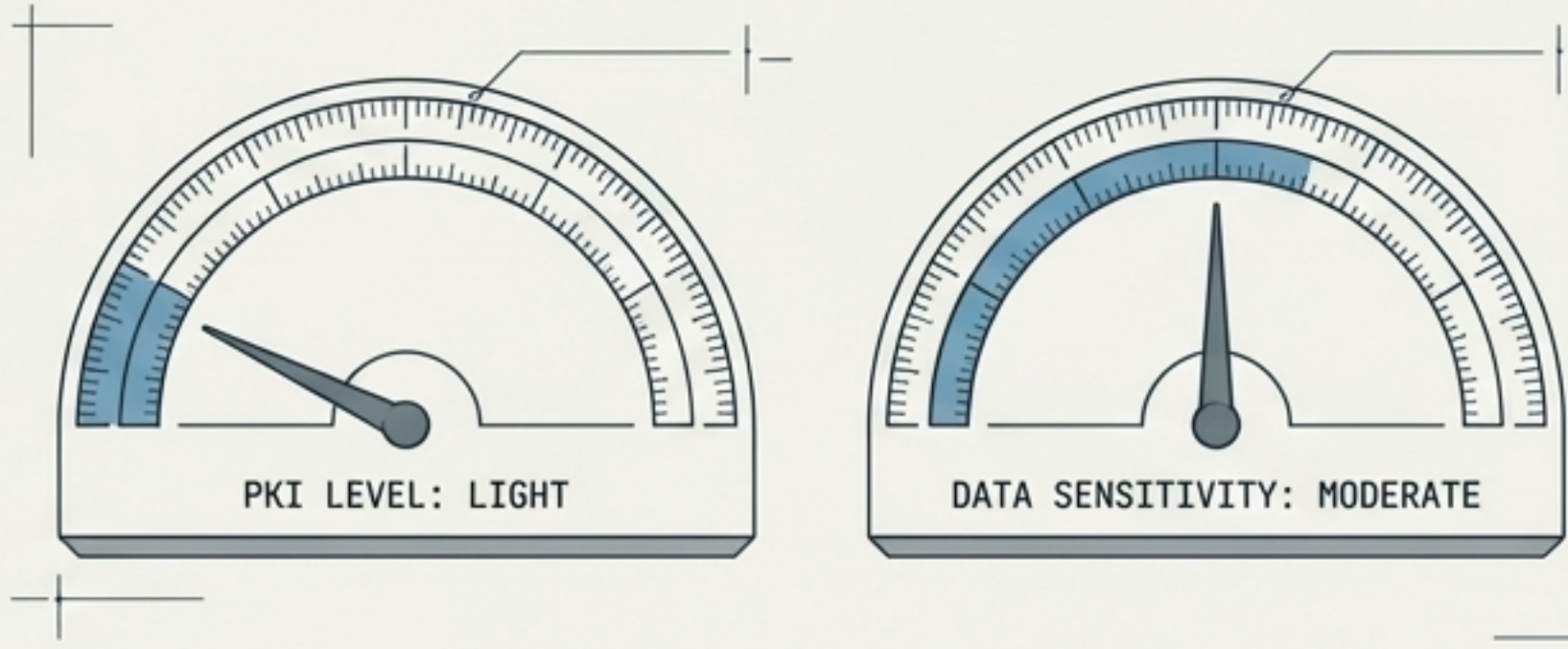
WORKFLOW



Sequential, one-directional, and often one-off.
Nobody is editing the same file at the same time.

[USE CASES: ACCOUNTANCY, MORTGAGE BROKERS, CONSULTING]

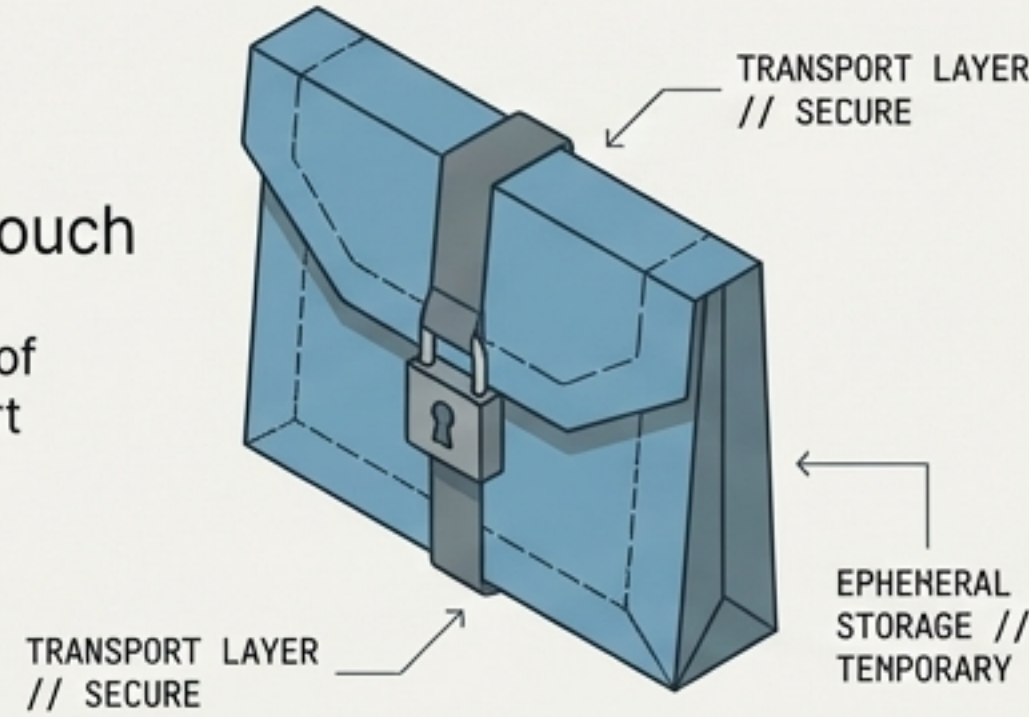
SECURITY GAUGES



VAULT METAPHOR

Vault Role:
Ephemeral Courier Pouch

The vault is not the source of truth. It is a secure transport layer replacing email attachments.



BUSINESS MODEL



The Commercial Opportunity

Serverless, low-cost operations.
White-label the SG/Send transfer
and add domain-specific features
(e.g., receipt categorisation).

“Hey Claude, here is my accountant's public key, send this tax information.”

Tier 2: Multi-Party Sequential Workflows

WORKFLOW



Hand-offs with defined boundaries.
Not everyone sees everything.

JetBrains Mono [USE CASES: SUPPLY CHAIN MANAGEMENT, BOARD GOVERNANCE, MULTI-PARTY NEGOTIATIONS]

SECURITY GAUGES

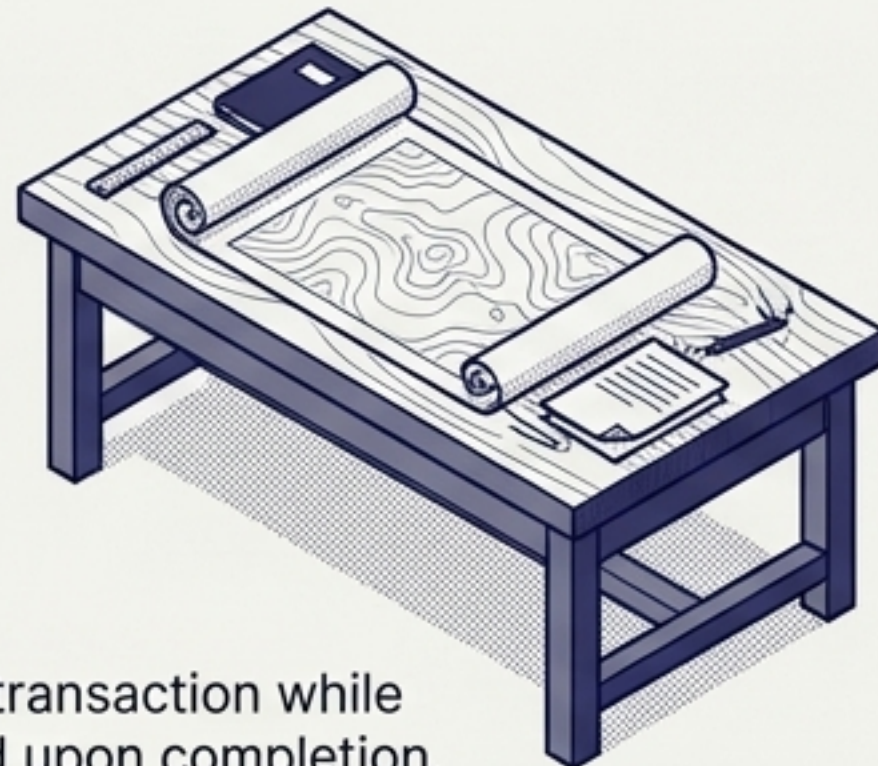


PKI LEVEL: MEDIUM



DATA SENSITIVITY:
MODERATE TO HIGH

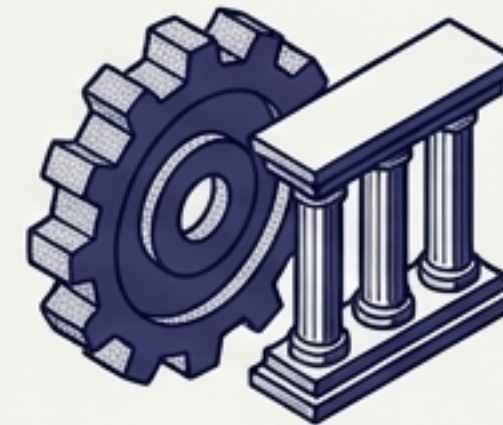
VAULT METAPHOR



Vault Role: Working
Source of Truth

The authoritative state of a transaction while in-flight. Archived or deleted upon completion.

BUSINESS MODEL

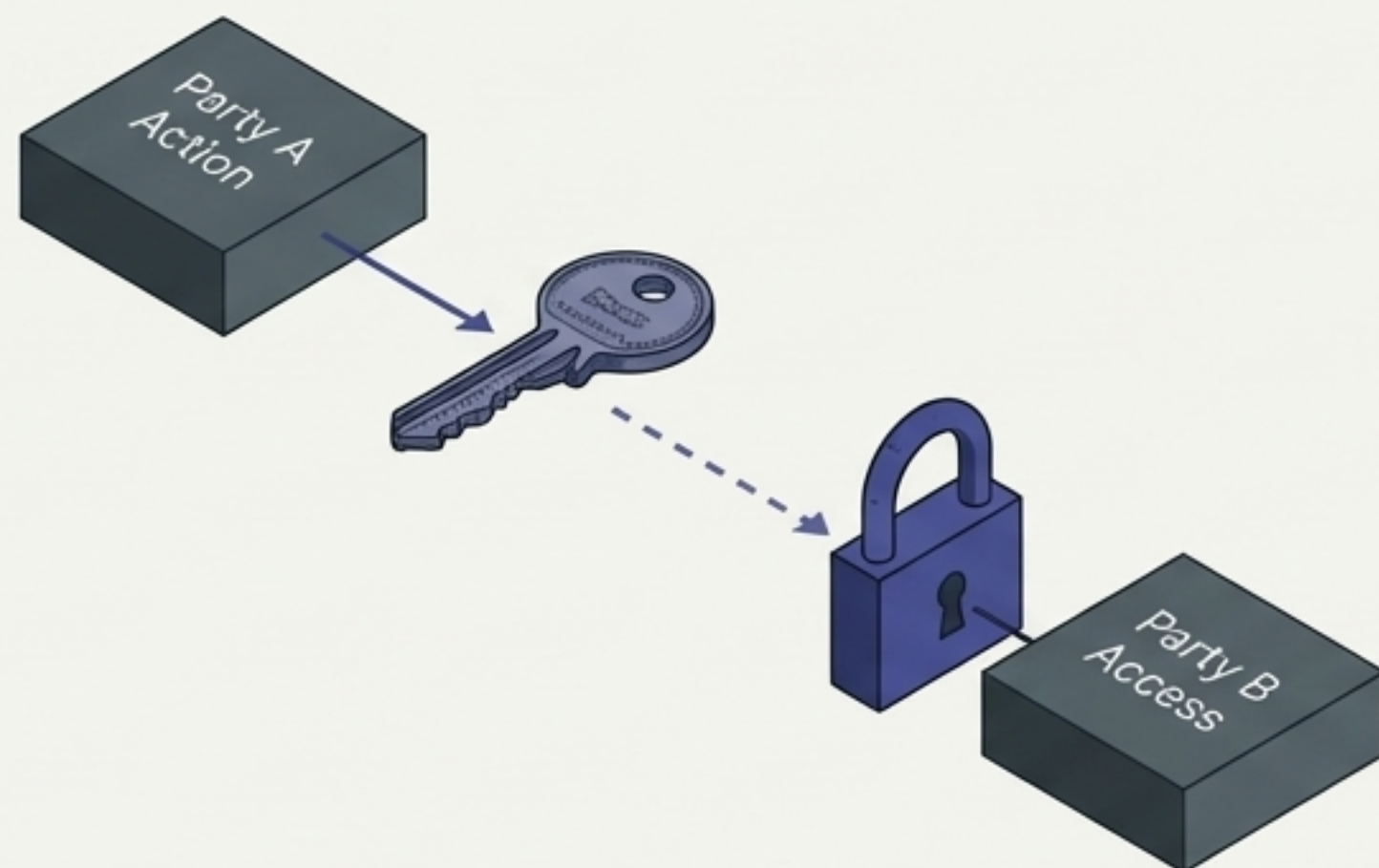


The Commercial Opportunity

The startup provides domain expertise and workflow UI; SG/Send provides the plumbing. Requires Town Planner architecture: backups, redundancy, and reliability guarantees.

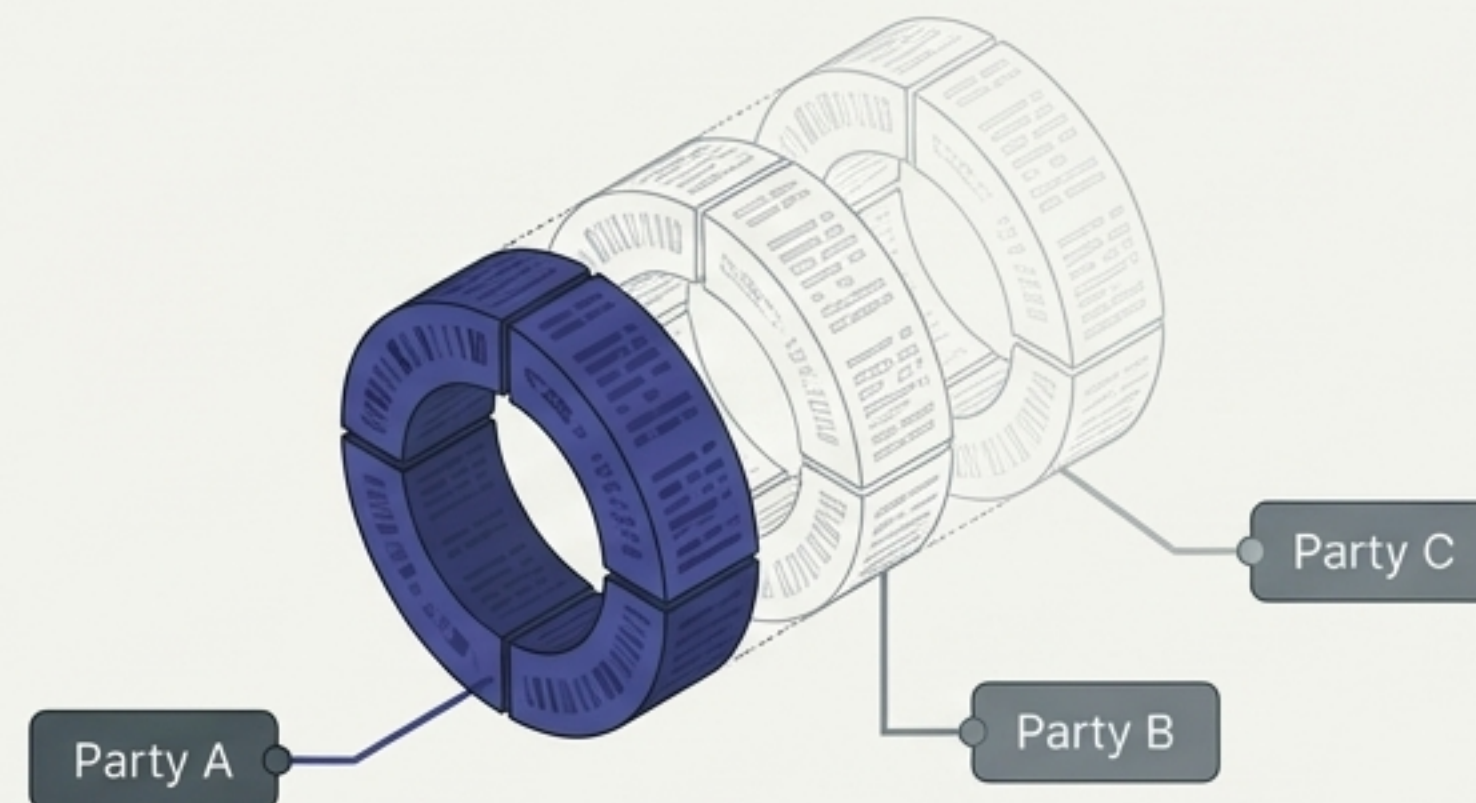
Trust Enforced by Maths, Not Policy

Sequential Enforcement



Sequential Enforcement: Party B physically cannot commence until Party A's cryptographic sign-off unlocks their state.

Cryptographic Slicing



Data Minimisation & GDPR: Per-party encryption. A compromise exposes only one slice. Server location is irrelevant (holds ciphertext). Right to erasure achieved instantly by deleting the vault key.

Tier 3: Agent-Driven Workflows

WORKFLOW



Automated systems coordinating work. Requires start/stop/replay debugging and branch-per-agent.

JetBrains Hone [USE CASES: LEGAL DOCUMENT PROCESSING, AI RESEARCH PIPELINES, AUTOMATED COMPLIANCE]

SECURITY GAUGES

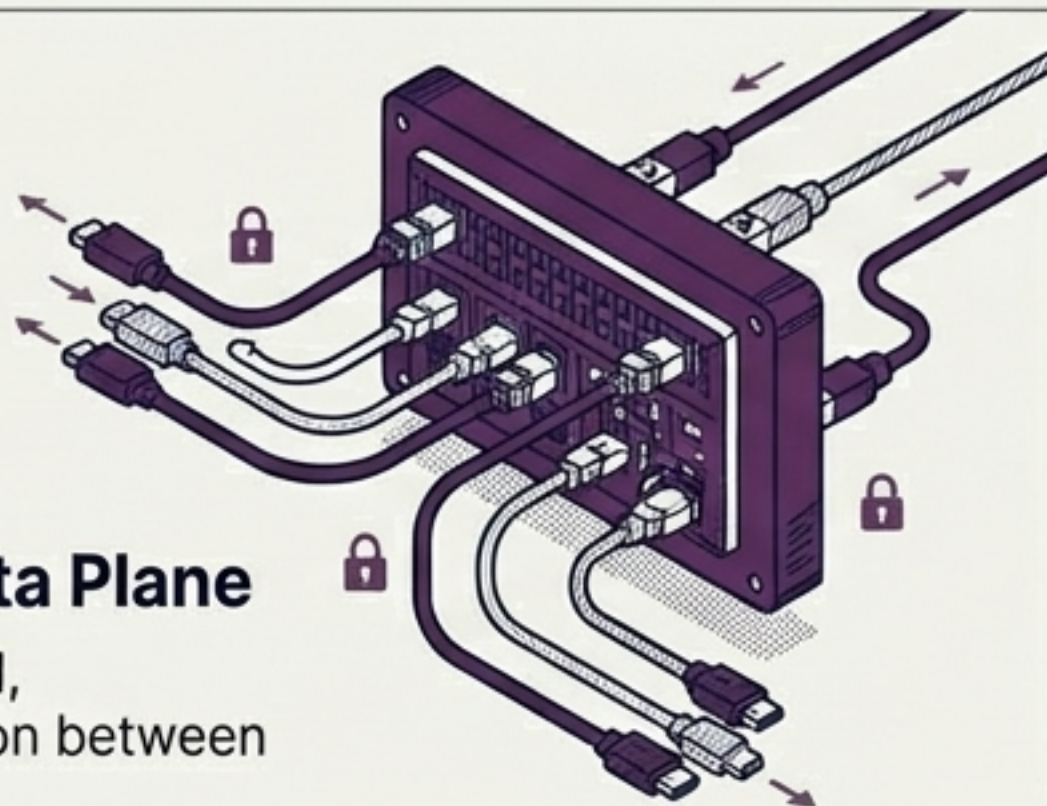


PKI LEVEL: MEDIUM-STRONG
(PER-AGENT)



DATA SENSITIVITY:
HIGH

VAULT METAPHOR



Vault Role: The Data Plane

Authenticated, encrypted, sequenced communication between automated entities.

BUSINESS MODEL

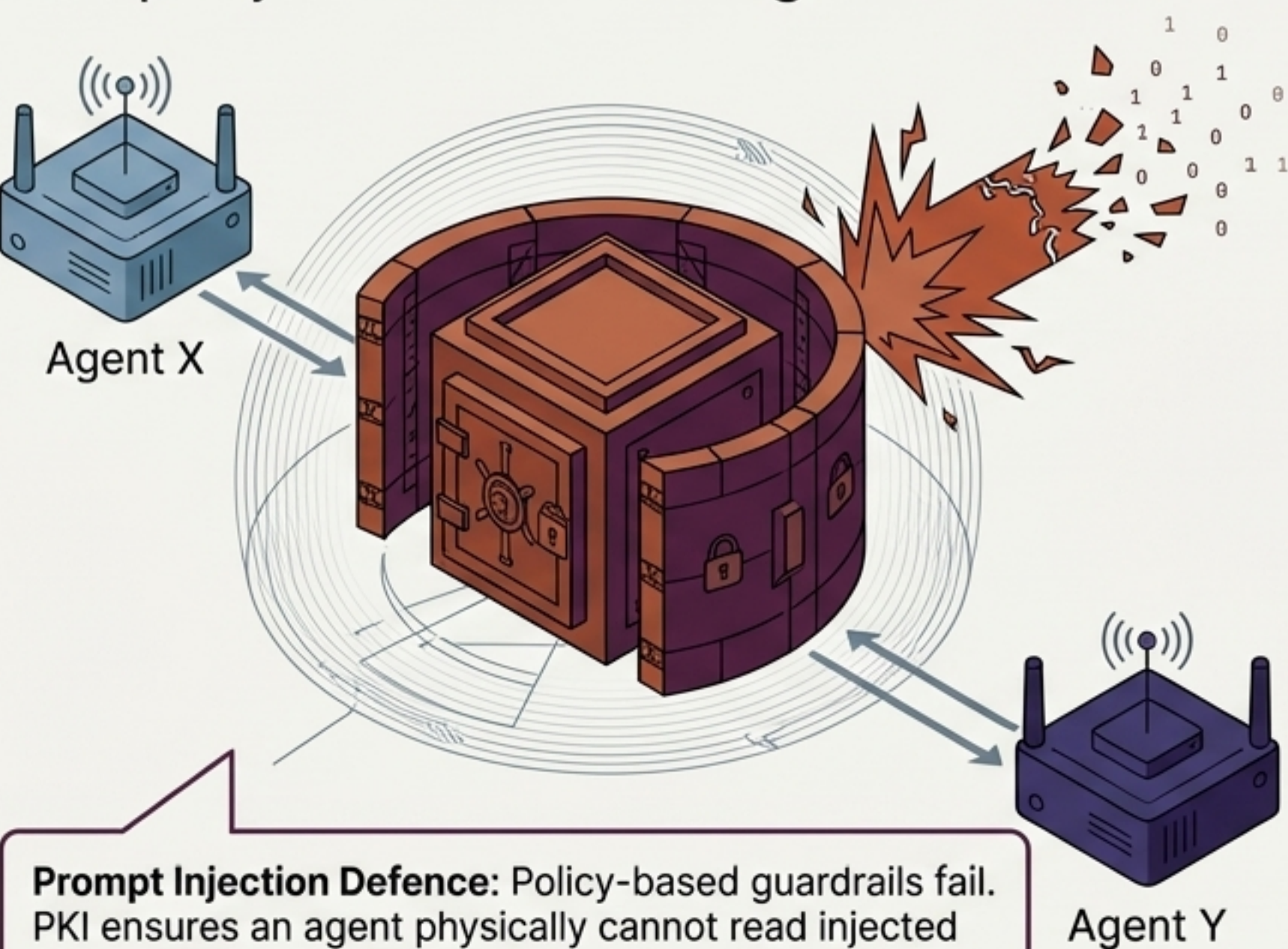


The Commercial Opportunity

Technically sophisticated startups. High value in customised Dev Packs (agentic team templates, role definitions, CI/CD setups).

Securing the Agentic Plane

Prompt Injection Defence Diagram



Prompt Injection Defence: Policy-based guardrails fail. PKI ensures an agent physically cannot read injected instructions meant for another without the private key.

Blockchain vs Vault Mini-Matrix

Traditional Blockchain	SG/Send Vault Model
Inter Public Ledger	Inter Signed & Encrypted
Inter Consensus Overhead	Inter Append-Only
Inter High Gas Costs	Inter Zero Consensus Overhead

Insight: The vault model delivers what most enterprise blockchain use cases actually needed.

Tier 4: High-Integrity, Signed Environments

WORKFLOW



Every action is signed. Every access logged.
Signatures assign legal responsibility, not just authentication.

JetBrains Rono Use Cases: Healthcare (patient records), legal (evidence), government filings.

SECURITY GAUGES



**PKI Level: Strong
(Mandatory Signing)**

JetBrains Mono



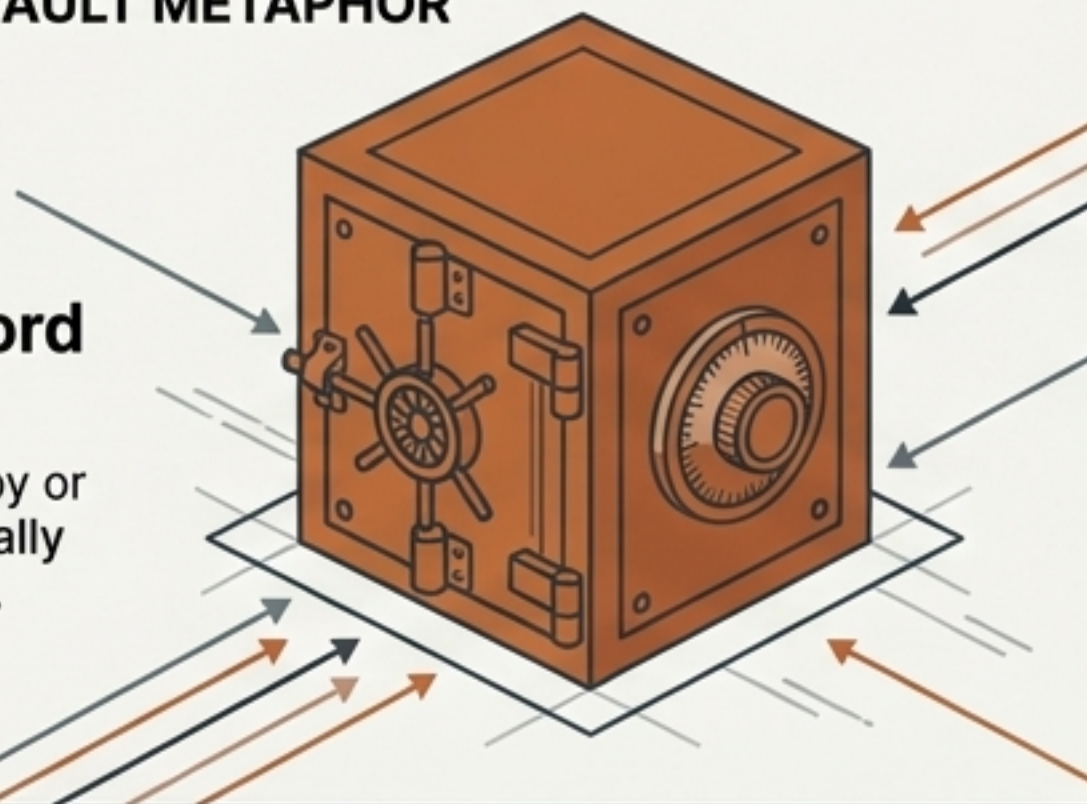
**Data Sensitivity:
Very High**

JetBrains Hono

VAULT METAPHOR

**Vault Role:
Authoritative Record**

The vault is no longer a copy or transport layer; it is the legally meaningful source of truth.



BUSINESS MODEL

**The Commercial
Opportunity**

Competing on SLAs, uptime guarantees, and audit certifications.

“Try proving cryptographically who accessed what and when using SharePoint.”



Tier 5: Network Isolation and Cross-Domain

WORKFLOW

JetBrains Mono



Networks isolated for security (unpatchable servers, industrial control systems) still need data. Opening ports defeats isolation.

Use Cases: Critical infrastructure, defence, manufacturing.

SECURITY GAUGES

JetBrains Mono



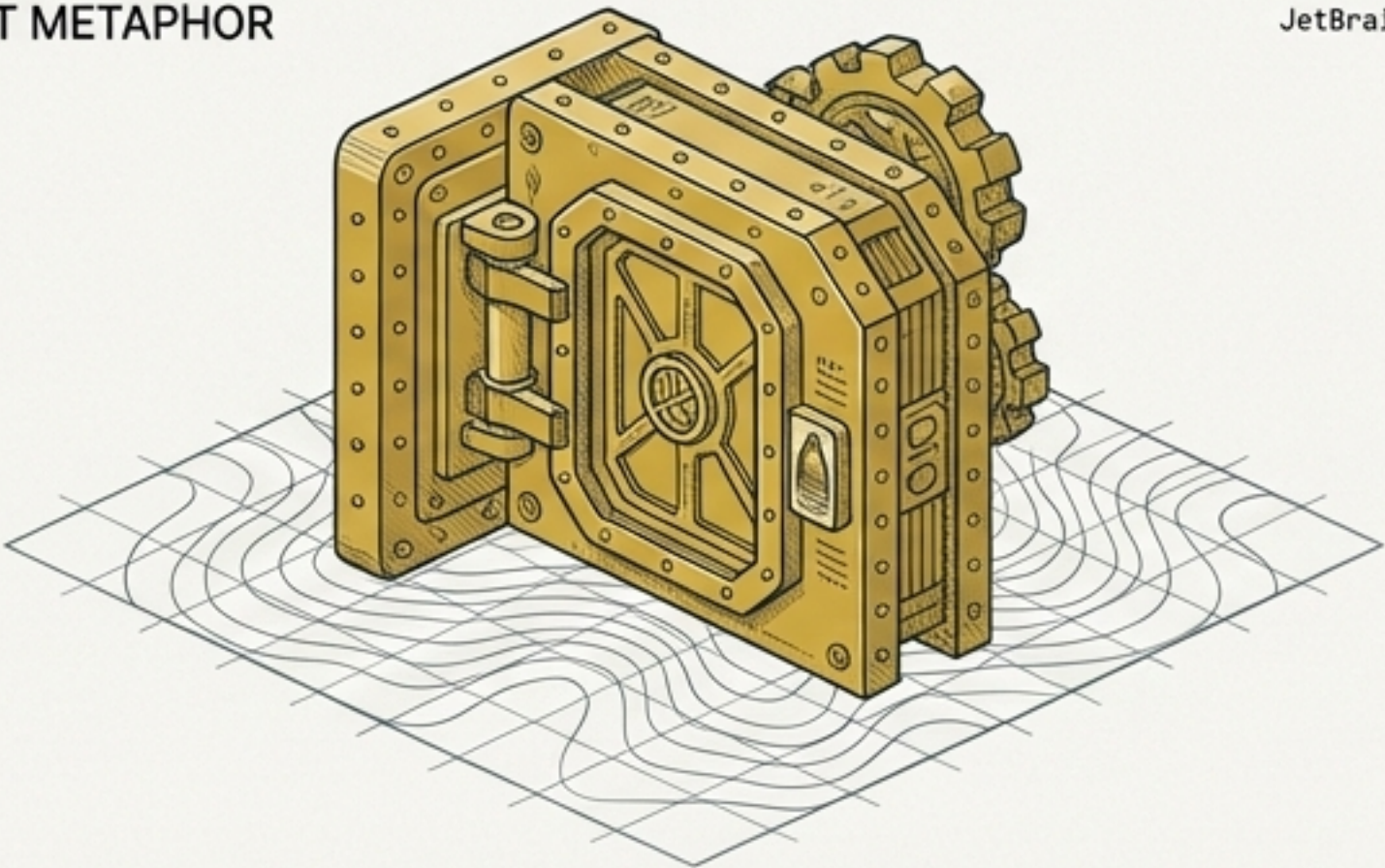
PKI Level: Strong
(Typed Data & Source Verification)



Data Sensitivity:
Extreme / Air-Gapped

VAULT METAPHOR

JetBrains Mono



BUSINESS MODEL

JetBrains Mono



Vault Role: Secure Data Bridge

Strongly typed data enters isolated networks via files (USB, one-way diode) without persistent open connections.

BUSINESS MODEL

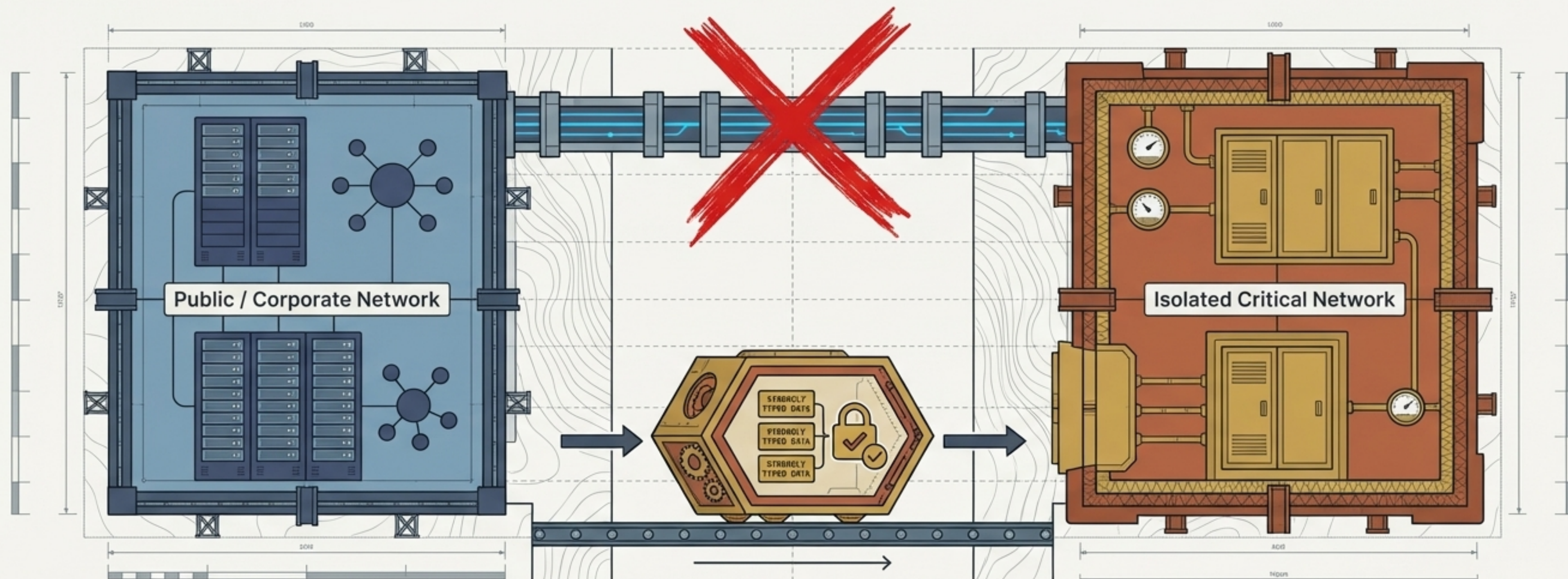
JetBrains Mono



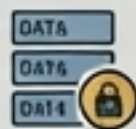
The Commercial Opportunity

Long sales cycles, high contract values. Requires security clearances and Common Criteria certifications.

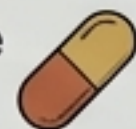
Bridging the Air-Gap Without Open Ports



1. Package, encrypt, and sign strongly typed data.



2. Transfer as a self-contained file (batch schedule, USB, diode).



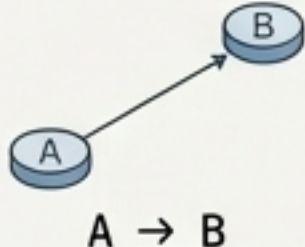
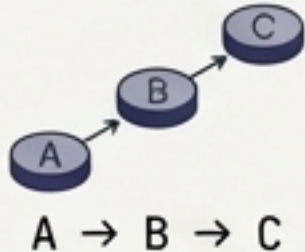
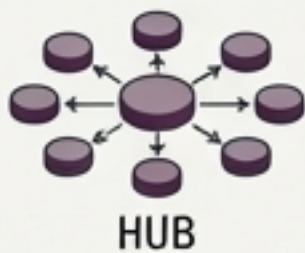
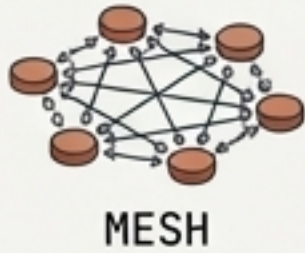
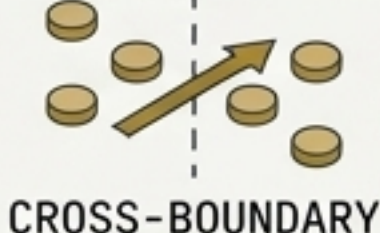
3. Receiving system decrypts and validates the signature.



Takeaway: Zero open ports. Zero persistent connections. Complete source verification.

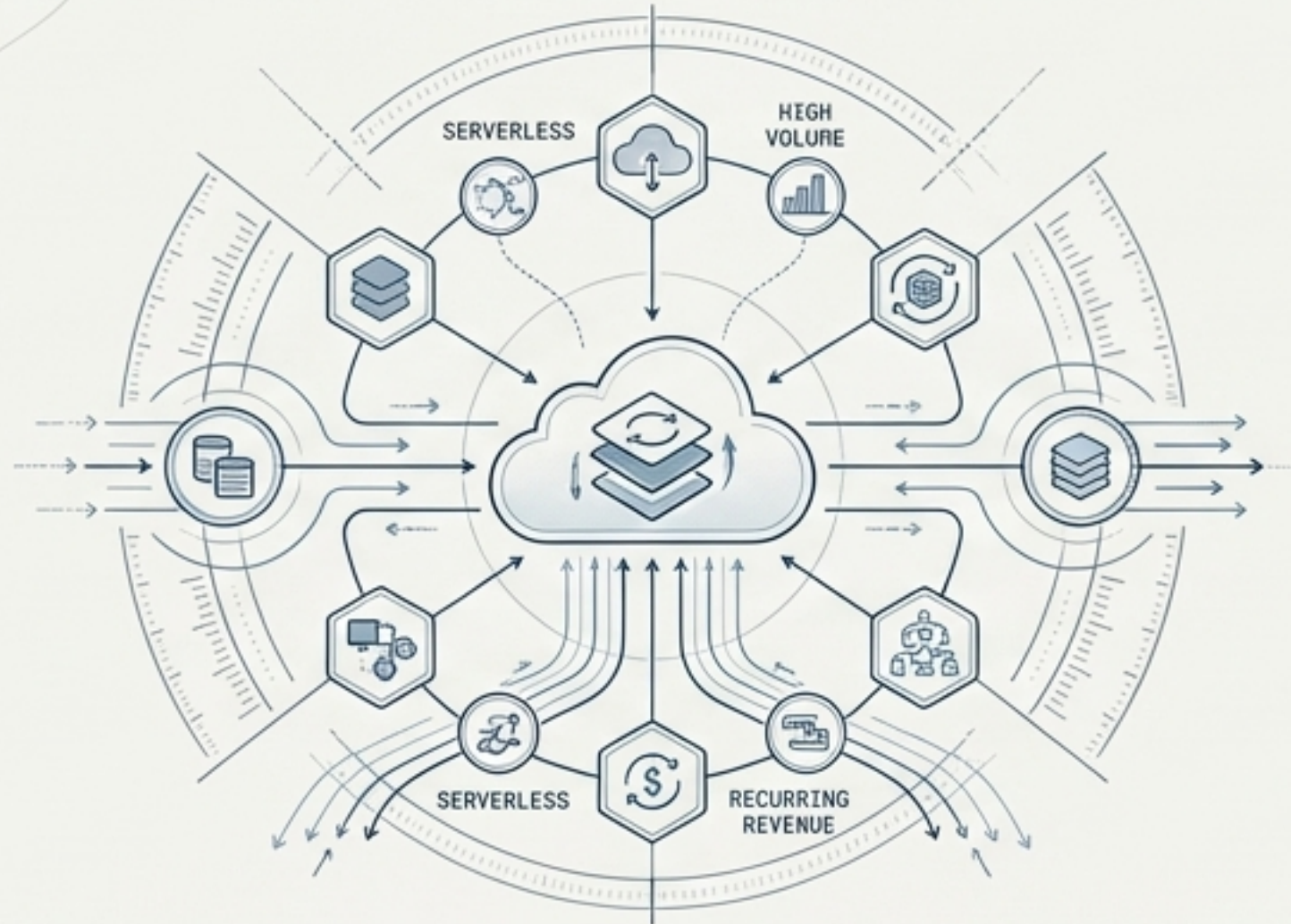


The Architecture of Trust: Diagnostic Toolkit

Tier	Vault Role	PKI Architecture	Collaboration Typology	Key Example
Tier 1	Ephemeral Transport	Light (shared keys)		Accountant sends tax docs
Tier 2	Working Source of Truth	Medium (per-party)		Supply chain order fulfilment
Tier 3	Data Plane	Medium-Strong (per-agent)		AI research pipeline
Tier 4	Authoritative Record	Strong (mandatory signing)		Clinical trials
Tier 5	Secure Data Bridge	Strong (source verification)		Air-gapped network updates

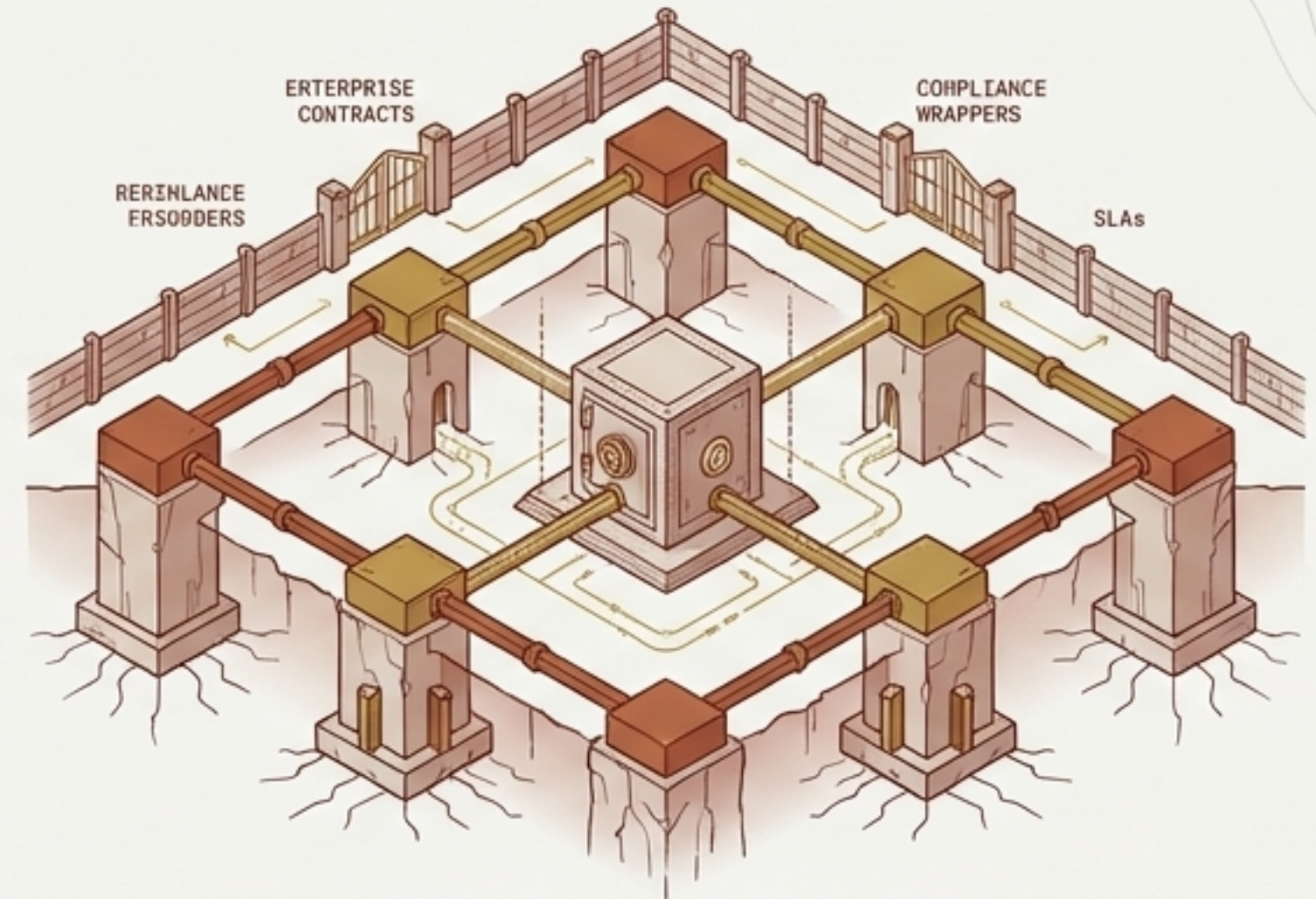
The Commercial Blueprint: Where Will You Build?

The Workflow Builder (Tiers 1 & 2)



Model: Serverless, high volume, recurring revenue.
Value Prop: Domain-specific UX and frictionless UX. SG/Send handles the encrypted infrastructure.

The Town Planner (Tiers 4 & 5)



Model: Enterprise contracts, compliance wrappers, SLAs.
Value Prop: Redundancy, disaster recovery, and cryptographic proof of provenance.

The primitives remain the same. The complexity and the commercial opportunity scale with the trust you provide.